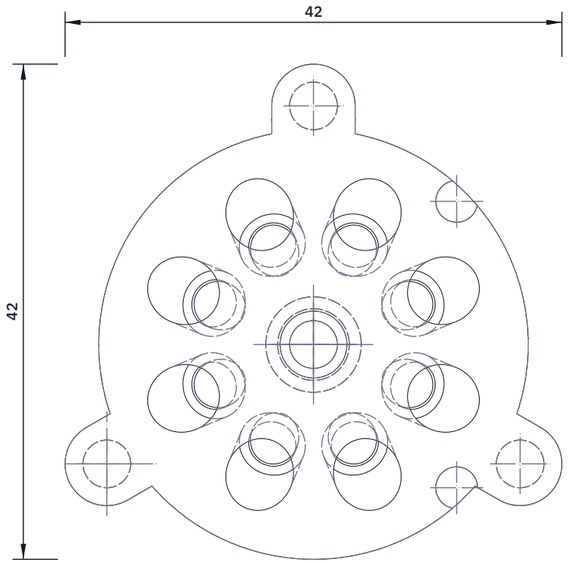
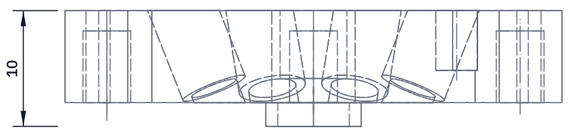
REV	DESCRIPTION	DATE	BY
P1	Making sketch for the prototype build plan	2026-10-24	AC



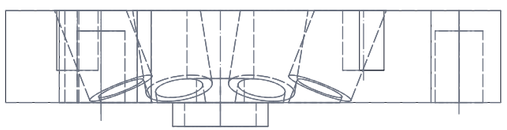
TOP VIEW

Scale 2:1; looking down (along -Z)



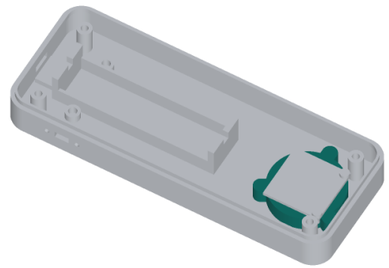
FRONT VIEW

Scale 2:1; looking at the front (along +Y)



RIGHT VIEW

Scale 2:1; looking at the right side (along -X)



WHERE IT GOES

This part in colour, its neighbours in grey

HOW TO MAKE IT AND HOW IT FITS

Print in black PETG, top face on the bed, 0.1 mm layers, 100 % infill.
Disc 36 mm across, 7.2 mm tall, with three lugs for the shroud screws on a 40 mm circle; press an M3 insert into each lug from below.
Eight LED pockets 5.6 mm on a 20 mm circle, each tilted 23 degrees inward so the beams meet on the item; a 4 mm light channel runs from each LED face out through the underside; leads come out of the top.
Centre bore 5.6 mm for the photodiode can, with a step 4 mm up; below it a 4 mm aperture, then a baffle tube 8 mm across standing 2 mm below the underside, through the window hole; its 6 mm end bore holds the 5.8 mm detector window flush with the glass face (fix F1).
Two 3.5 mm holes 12 mm each side for M2.5 inserts (amplifier board).
Ream the LED pockets and the bore with a 5.6 mm drill by hand.
Check: each LED can slides in and stops square; the bore is black inside.

WASTEWISE SCAN github.com/BoujeeEnjinia1701/wastewise-scan

WasteWise Scan optical head block: making sketch

DRAWING NO.	REV	SHEET	SIZE
WSC-DWG-103	P1	1 of 1	ANSI B

SCALE	UNITS	PROJECTION	DATE	DRAWN
2:1	mm		2026-10-24	Amish Chadha

MATERIAL / NOTES
Black PETG, 3D printed

Licensed CERN-OHL-S-2.0 · Design Molecule Lab · designmolecule.com

Generated from cad/src; do not edit by hand